

6. A class investigated the rate of the reaction between magnesium and hydrochloric acid by measuring the volume of gas produced in 10 seconds using different concentrations of acid.

- (a) Tick (✓) the name of the piece of apparatus they used to measure the volume of gas produced. [1]

beaker

☐

thermometer

☐

gas syringe

☐

conical flask

☐

- (b) The table shows the results of their experiment.

| Concentration of acid (M) | Volume of gas produced (cm <sup>3</sup> ) |        |        |      |
|---------------------------|-------------------------------------------|--------|--------|------|
|                           | Test 1                                    | Test 2 | Test 3 | Mean |
| 0.2                       | 16                                        | 14     | 15     | 15   |
| 0.4                       | 31                                        | 33     | 30     | 32   |
| 0.6                       | 47                                        | 49     | 29     | 48   |
| 0.8                       | 63                                        | 64     | 65     | 64   |
| 1.0                       | 82                                        | 83     | 79     | 81   |

- (i) **Circle** the result in the table which was **not** used to calculate a mean.

Give the reason why this result was not used.

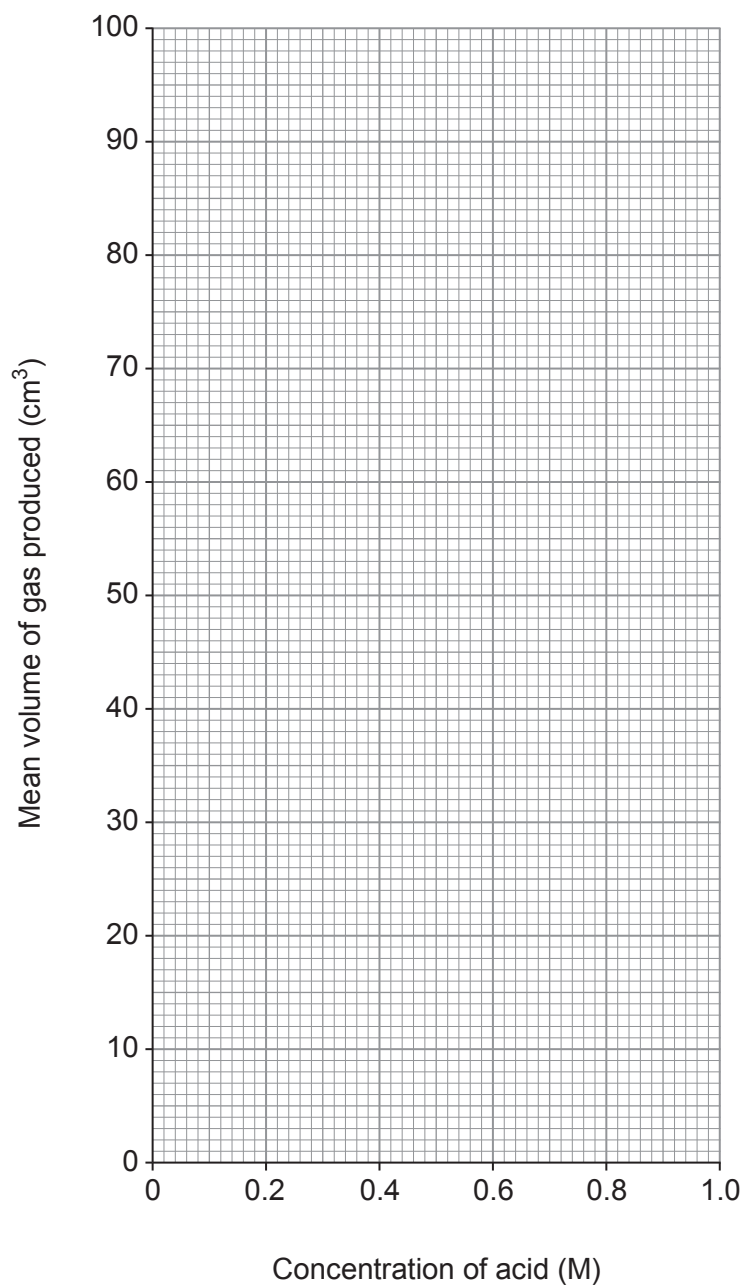
[2]

Reason .....

.....



- (ii) Plot the mean volume of gas produced against the concentration of acid. Draw a suitable line. [3]



- (iii) Use the information to find the volume of gas produced in 10 seconds using acid of concentration 0.5M. [1]

..... cm<sup>3</sup>



Examiner  
only

(c) In another investigation the class found that as the temperature of the acid increases the reaction is faster.

Tick (✓) the **two** statements which explain why the reaction is faster at a higher temperature.

[2]

- there are more particles of acid☐
- the acid particles are moving faster☐
- there is a higher surface area☐
- there are more collisions per second☐
- the acid particles have less energy☐

(d) The class wanted to identify the gas produced. They tested it as shown in the table.

|             |                                  |                                       |
|-------------|----------------------------------|---------------------------------------|
| Test        | bubble the gas through limewater | place a burning splint in the gas     |
| Observation | limewater did not change         | a squeaky pop and the splint went out |

State the name of the gas.

[1]

.....

10

